

Lab for Difference in Difference Designs

Chapter 13

Case Study

Museum Science Spatial: A Program for Pre-Schoolers

A STEM Museum in Florida has had a longstanding partnership with two researchers at a nearby university. Drs. Andrews and Zheng are interested in how informal experiences in museums may encourage families to engage with their children in ways that inspire and support STEM learning at home. This interest has led them to work with the museum staff to design a series of exhibits that foster interactions (while at the museum), while also providing materials to take home, with the hope of continued engagement. Understanding if this program actually improves these connections is important, given the continued cost of these take-home materials. This study focuses on a particular exhibit that focuses on fostering the development of spatial skills through a series of building activities.

Plausible Evaluation Design

Difference in Difference Analysis

Suppose researchers wish to evaluate the effectiveness of a similar program on parent interaction with their children at home, measured using a well-established validated scale based on questions answered through an online survey collection system. The design is to recruit families as they enter the museum into the study, as them to fill out a baseline survey, then randomly assign them to attend the additional exhibits and receive the additional take home materials (treatment group) or attend the typical exhibits (control group). After 30 days, all study families take a survey and their answers for the baseline and follow-up surveys are used to compute 2 outcome scores for each family, one for baseline and one for followup.

For simplicity, assume perfect compliance with random assignment and a perfect response rate.

Question 1

Using the planner tool, find the minimum detectable effect for a study in which 20 families are assigned to the treatment group and 20 families are assigned the control group when power is .8 for a test with a significance level of .05. For this question, assume 0 correlation between baseline and followup surveys.

```
compute_mdes(V = did_fixed_nocovariate_V(nt = 20,
                                           nc = 20,
                                           ar = 0),
             df=srs_fixed_nocovariate_df(nt = 20,
                                           nc = 20),
             beta = 1-.8,
             alpha = .05
)
```

```
## [1] 1.28542
```

Question 2

A member of the study team claims that a better study would use a higher portion of the study families as treatment. Using the planner tool, find the minimum detectable effect for a study in which 30 families are assigned to the treatment group and 10 families are assigned the control group when power is .8 for a test with a significance level of .05. For this question, assume 0 correlation between baseline and followup surveys.

Is this design more sensitive?

```
compute_mdes(V = did_fixed_nocovariate_V(nt = 30,
                                           nc = 10,
                                           ar = 0),
             df = did_fixed_nocovariate_df(nt = 30,
                                           nc = 10),
             beta = 1-.8,
             alpha = .05
            )
```

```
## [1] 1.484275
```

Question 3

There is some concern that the study may not be as sensitive as the researchers need it to be. The **first idea** is to use 60 families, with 30 in treatment and 30 in control, and continue to assume that the baselines and followups are not correlated. The **second idea** is to keep the 40 study members (20 in each group), but assume that the baseline and followups are likely to be correlated at $\rho = 0.5$.

Using the planner tool, find the minimum detectable effect for a study in which 30 families are assigned to the treatment group and 30 families are assigned the control group, assuming no correlation between baseline and followup surveys, when power is .8 for a test with a significance level of .05. This is the sensitivity of the **first idea**.

```
compute_mdes(V = did_fixed_nocovariate_V(nt = 30,
                                           nc = 30,
                                           ar = 0),
             df = did_fixed_nocovariate_df(nt = 30,
                                           nc = 30),
             beta = 1-.8,
             alpha = .05
            )
```

```
## [1] 1.040139
```

Using the planner tool, again, find the minimum detectable effect for a study in which 20 families are assigned to the treatment group and 20 families are assigned the control group, assuming the correlation between baseline and followup is $\rho = 0.5$, when power is .8 for a test with a significance level of .05. This is the sensitivity of the **second idea**.

```
compute_mdes(V = did_fixed_nocovariate_V(nt = 20,
                                           nc = 20,
                                           ar = .5),
             df = did_fixed_nocovariate_df(nt = 20,
                                           nc = 20),
             beta = 1-.8,
             alpha = .05
            )
```

```
## [1] 0.9089294
```

Which idea is more sensitive?

Question 4

Questions 1-3 involve the same sample parameters as the lab for Chapter 7 (questions A-1 through A-3). Compare the results of this lab to your answers for that first lab.

References

Marcus, M., Haden, C.A., & Uttal, D.H. (2017) STEM Learning and transfer in a children's museum and beyond. *Merrill-Palmer Quarterly*, 63(2): 155-180.